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RESEARCH ARTICLE-COMPUTER ENGINEERING AND COMPUTER SCIENCE

Drone Applications in Logistics and Supply Chain Management: A Systematic Review Using Latent Dirichlet Allocation

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Abstract
As an emergent technology, drones have grown increasingly integral to the logistics and supply chain management (SCM) sectors. This review paper seeks to address this gap by analyzing a carefully curated dataset of 96 journal articles, sourced from Scopus and Web of Science, concerning drone applications in logistics and SCM. Our study reveals a rising trend in drone-related logistics and SCM research and highlights the growing significance of this technology. Using latent Dirichlet allocation (LDA), we distilled the corpus into ten distinct topics and provided a comprehensive overview of the field's thematic landscape. The most prominently studied themes were drones for humanitarian logistics and SCM and optimization of last-mile delivery using drones and trucks in urban logistics. These areas underscore the promise of drones in addressing humanitarian challenges and urban logistics optimization, respectively. Conversely, the least explored topics were drones safety and utility and drone delivery and consumer behavior in food logistics. To our knowledge, this is the first paper applying LDA to examine drone applications in logistics and SCM, offering novel insights into the current state of the literature. Moreover, based on our findings, we propose a future research agenda to encourage further exploration into under-studied areas and to sustain the momentum of this exciting field.

Keywords Drones · Logistics · Supply chain management · Humanitarian logistics · Last-Mile delivery

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1 Introduction

The emergence of drones, or unmanned aerial vehicles, presents a promising solution to bring about a new era of innovation in logistics and supply chain management (SCM) [1]. With their unique capabilities and advanced technologies [2], drones can help overcome these challenges associated with conventional methods and systems. Rejeb et al. [1] indicate that drones can be employed for various logistics and SCM applications, including last-mile delivery, warehouse management, and inventory tracking. Drones can deliver packages directly to customers' locations, enhance delivery speed, and reduce the need for extensive ground transportation infrastructure [3]. By circumventing traffic congestion and other obstacles, drones can remarkably reduce delivery times, minimize operational costs, and foster environmental sustainability [4]. Furthermore, drones with advanced sensors can streamline warehouse operations by automating inventory monitoring and management. With the support of drones in tasks such as RFID tags or barcodes, organizations can minimize the risk of human error, expedite

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inventory management, improve the accuracy of data, and enable effective decision-making processes. As a result, the technology can increase cost savings and contribute to a more agile and responsive supply chain [5]. In addition, drones can be used to monitor logistics and SCM operations in real time and provide valuable data that can be leveraged to support decision-making and mitigate risks. Through aerial surveillance and data collection, drones can monitor the condition of infrastructure, detect anomalies or potential bottlenecks in the supply chain, and ensure the timely flow of products and information [1]. Therefore, managers can capitalize on this real-time visibility to make informed decisions, address potential issues proactively, and maintain optimal performance across the entire supply chain.

While the importance of drones in logistics and SCM has spurred considerable academic research in recent years, the existing literature reviews on this subject have primarily employed traditional systematic review methods. For example, Rejeb et al. [1] review the potentials and challenges of drones in logistics and SCM using 55 articles. The findings indicate that the benefits of drones include support of humanitarian logistics, reduced delivery time, cost efficiency, higher flexibility, and increased sustainability, while the challenges of the technology are grouped into organizational, technical, regulatory, and safety-related issues. Mohammad et al. [6] conduct a comprehensive review and analysis of the latest trends in last-mile delivery solutions from both industry and academic perspectives by analyzing over 80 relevant articles. The authors highlight critical logistic challenges raised by the surge of online retail activities and propose various innovative substitutes to conventional truck-based home delivery, including delivery by drones. In the humanitarian logistics context, Rejeb et al. [7] comprehensively explore the potential capabilities, performance outcomes, and limitations of drones by analyzing 142 articles through a systematic literature review to develop a comprehensive framework for their potential applications.

Furthermore, the review of Moshref-Javadi and Winkenbach [8] offers a framework that classifies drone-based delivery systems and underscores gaps in research, emphasizing the significant attention given to single-truck, single-drone models in Synchronized Multi-modal delivery. This research indicates that areas like resupply multi-modal models and applications such as food and mail deliveries are still relatively untapped. Kellermann et al. [9] shed light on the evolving conversation surrounding delivery drones and air taxis. The authors identify predominant technical barriers and economic potentials while advocating for robust evidence supporting societal benefits and encouraging broader public participation. Chung et al. [10] dive into the optimization methodologies used in drone and drone–truck operations across diverse sectors. The authors elaborate on mathematical models, synchronization techniques, and implementation hurdles, while also spotlighting existing research voids and proposing avenues for future exploration. In another review, Merkert and Bushell [11] elucidate a shift in focus from concerns centered on privacy to those related to operational hurdles. The authors underscore an urgent need for improved policies, like the introduction of low altitude airspace management systems, to harmonize drone expansion with innovation. Abdelmaboud [12] delves into the increasing integration of drones in everyday life, emphasizing the burgeoning trend of the Internet of Drones (IoD) aimed at bolstering flight safety and quality. His study provides insights into pivotal security, privacy, and communication prerequisites, showcases a taxonomy rooted in relevant parameters, highlights notable commercial case studies, and ponders the challenges and future research prospects for IoD.

Similarly, Zhang et al. [13] take a discerning look at the interplay between trucks and drones in the delivery ecosystem. Their analysis puts forth a taxonomy, engages in discussions about models and algorithms, and accentuates present-day challenges along with forthcoming trajectories in logistics research. Madani and Ndiaye [14] focus on the potential and challenges presented by hybrid truck–drone systems for last-mile delivery, stressing aspects like on-the-fly optimization, the ecological footprint, and air traffic administration. The research of Benarbia and Kyamakya [15] delves deep into the ascendancy of drones as an answer to the challenges that e-commerce enterprises encounter during last-mile deliveries. The study offers a sweeping survey on drone logistics, zeroes in on crucial solutions, and envisages the future of drone-based delivery systems. In the same vein, Li et al. [16] examine the pivotal role drones play in resolving logistics challenges driven by e-commerce. Their analysis revisits recent research on UAV applications in logistics, organizing findings into categories like models, applications, and additional focal points while envisioning subsequent research pathways. Lastly, Rojas Viloria et al. [17] review academic contributions on drone routing problems spanning from 2005 to 2019, organizing the results by objectives, solution methods, applications, and constraints, and culminating with insights into emerging trends and future directions.

Collectively, the prior reviews have provided valuable insights into the potential benefits, challenges, and implications of drone integration in logistics and SCM. However, they have also faced certain limitations, including the potential for subjective bias in the selection and interpretation of studies, as well as the difficulty of keeping up with the rapidly evolving body of knowledge in this domain. While there have been only a few attempts to review the literature surrounding drone applications in logistics and SCM, none of these prior works has utilized the Latent Dirichlet Allocation (LDA) technique. As a topic modeling approach, LDA offers several advantages over traditional systematic reviews, including the

ability to analyze large volumes of text data in an unsupervised manner, to identify latent semantic structures within the literature, and to uncover previously unexplored relationships among studies [18]. The absence of LDA-based reviews in the context of drone applications in logistics and SCM highlights a knowledge gap in the current literature. By employing the LDA technique, researchers can uncover new patterns, trends, and knowledge clusters that may have been overlooked in previous reviews, thereby contributing to a more comprehensive and nuanced understanding of the field.

The rest of this article is structured as follows: Sect. 2 outlines the methodology employed in this research. Section 3 provides an overview of the review's outcomes. Section 4 examines the topics identified using the LDA approach, and future research directions are also discussed. Finally, the article concludes by highlighting the implications and limitations of the research.

2 Methodology

In this study, we use the DLA technique, which is a hierarchical Bayesian model with three levels [19]. Theoretically, LDA is a series of algorithms for discovering and categorizing large text collections according to their topics. It is also a probabilistic generative tool for latent topic modeling [20]. The premise of LDA is that texts may be interpreted as probabilistically blended over a group of latent topics, each of which is defined by a unique word distribution [19]. LDA allows for the generation of K topics represented as βK, which is a probability distribution across terms in a collection of documents containing words from a lexicon V. This can be particularly useful for analyzing and categorizing large bodies of text data, allowing for efficient and automated extraction of meaningful themes and patterns. Based on the findings of previous studies [21, 22], we opted to use LDA for this study since it enables a systematic and objective evaluation of a huge body of literature. All the necessary statistical computation, graphical design, and natural language processing (NLP) activities were carried out in the software R and Python.

2.1 Article Selection

Scopus and Web of Science were consulted in order to assemble pertinent drone-related articles since they are extensively related to academic studies and can provide high-quality material in response to search queries. On May 1, 2023, we searched these two databases for the following keywords: (drone* OR "unmanned aerial vehicle*" OR uav* OR "unmanned aircraft system*" OR uas* OR "remotely piloted aircraft*") AND ("supply chain*" OR logistic* OR procurement OR purchasing OR sourcing OR manufacturing OR production OR inventory OR "storage assessment" OR "order picking" OR transport OR warehous*). While the OR operator is used to expand the search query by allowing the search engine to find any of the related terms, the AND operator is used to narrow down the search results by requiring that both sets of search terms are present in the documents being searched. The asterisk (*) is also used as a wildcard character to match any characters or words that follow a specific pattern. For example, the term "drone*" with an asterisk at the end can match any word that begins with "drone," such as "drones," "drone technology," "drone delivery," etc. Consistent with prior studies [23, 24], we solely included English-language articles in this investigation to guarantee the scholarly quality of the retrieved information. Table 1 provides a concise overview of our search process across the Scopus and Web of Science databases, detailing the keywords used and the specific criteria applied.

2.2 Corpus Generation

Before conducting the unsupervised LDA analysis, the collected texts must be prepared. URLs, new line characters, punctuation, and other symbols were stripped out in the first phase. The text was then altered by Gensim, an open-source Python model, by removing adverbs, adjectives, verbs, and stop words. To improve the accuracy of the analysis, some of the stop words from the Gensim collection were combined with other irrelevant terms. Finally, Gensim software was used to extract individual words from phrases and assign them unique identifiers (IDs). By adhering to these procedures, not only can the occurrences of individual words in the texts be counted, but their relative importance can also be evaluated.

2.3 LDA Model Development and Optimal K Number of Topics Identification

Creating an effective vocabulary is crucial for building the LDA model and extracting topics from the texts. The Gensim package's "id2word" function allows analysts to quickly and easily produce a vectorized bag of words. Mallet was then used to construct the LDA model [25]. Mallet is a versatile toolset with clustering, topic modeling, and document classification capabilities that can be used to analyze and gain insights from datasets. Various settings are required for the modeling process. To conduct this study, several simulated LDA models with varying numbers of topics were run via the Mallet software. The topic coherence score was used to determine the optimum number of topics. The role of coherence scores is to assess the cohesion of a topic by analyzing the relationships between its words. Using this metric, we assigned each topic a numerical score that reflects how well

Table 1 Database search process

Database	Search date	Keywords used	Criteria
Scopus	May 1, 2023	drone*, "unmanned aerial vehicle*", uav*, "unmanned aircraft system*", uas*, "remotely piloted aircraft*" "supply chain*", logistic*, procurement, purchasing, sourcing, manufacturing, production, inventory, "storage assessment", "order picking", transport, warehous*	English language Peer-reviewed articles
Web of Science	May 1, 2023	[Same as above]	[Same as above]

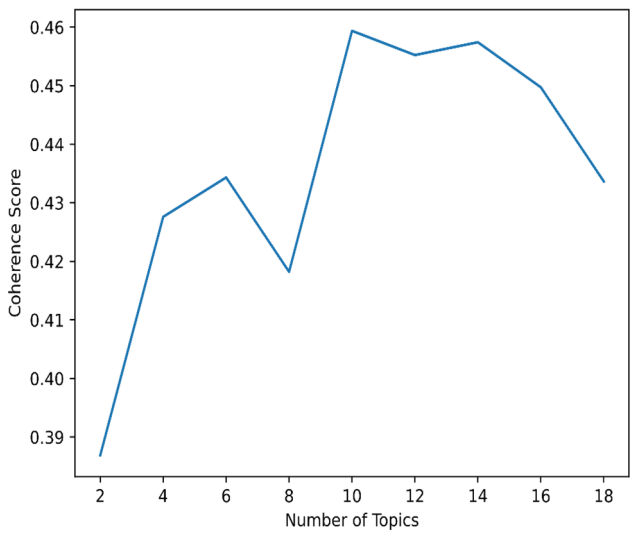
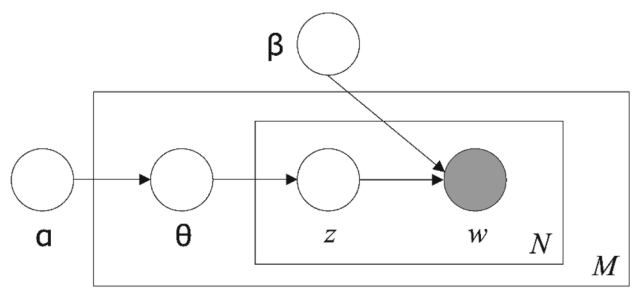


Fig. 1 Coherence scores plot

Table 2 Coherence scores per number of topics

Number of topics	Coherence score
2	0.3869
4	0.4276
6	0.4343
8	0.4182
10	0.4594
12	0.4552
14	0.4574
16	0.4497
18	0.4337



$$p(\theta, \mathbf{z}, \mathbf{w} \mid \alpha, \beta) = p(\theta \mid \alpha) \prod_{n=1}^{N} p(z_n \mid \theta) p(w_n \mid z_n, \beta)$$

Fig. 2 Representation of the LDA model [19]

its individual words work together to form a topic. Since a greater coherence score shows that the words within a topic are more cohesively related and substantially contribute to the topic, it is often preferred for precise and accurate topic modeling. Figure 1 displays the coherence scores generated by the unsupervised learning system. According to the concept of coherence score, the best LDA model would be the one with the highest and most stable score, showing more semantic coherence among the words inside each topic. The results show that after processing ten topics, the model settles down to a mean value of 0.4597. That is to say, adding more than ten topics to the research would not provide any useful results. Therefore, the model with ten topics was chosen as the best option for analysis based on the coherence scores provided by the LDA modeling procedure. Table 2 displays the coherence rates for all tested topic numbers.

2.4 Topics Identification

The LDA model represents a type of probabilistic model used to discover topics in a set of documents [20]. Figure 2 depicts the rectangles as replicates, M as documents, and N as the frequency of a topic in each document. The way in which the observed words, denoted as "w," are distributed is based on the topic distribution, denoted as "z." The distribution of words across topics is represented by "β," the distribution of topics across documents is represented by θ, and the distribution of words across topics is represented by α. The LDA model was applied to identify the frequency with which the ten topics in the literature were discussed in the selected journal articles. The semantic coherence approach was used by the researchers to count how many times topic-related terms occurred in the full text of each article. Two members of

the research team utilized an inductive method based on the semantic coherence score to select a set of articles for each topic. Ultimately, the LDA model enabled the detection of latent topics that were unique to each document, as well as their significance and frequency across texts, illuminating the current status of drone research in the logistics and SCM context. The LDA model that was constructed from our data was examined using various Python packages. The average distance between topics and the ten most important terms in our dataset was determined using PyLDAvis. Furthermore, the Matplotlib library was used to visualize the research findings.

2.5 Bibliometric Analysis

A bibliometric analysis was performed to learn more about the value of selected journal articles. The research team used the bibliometric R package to implement the methodology suggested by Aria and Cuccurullo [26]. This program makes it easier to find connections between scholarly publications, letting scholars get a deep comprehension of the data's underlying networks and topics. The study's principal aims, performance analysis and scientific mapping, were accomplished with the use of bibliometric techniques. Caputo et al. [27] explain how the former allows for an in-depth analysis of academic organization and researcher performance, while the latter aids scholars in understanding the origins and evolution of a certain field of study.

3 Findings

3.1 Descriptive Analysis

Table 3 presents relevant data extracted from the chosen articles included in this study. An important observation from the findings is that, on average, each article received 27.39 citations per year. This indicates a high level of interest and engagement from the research community, suggesting that the topic of drone applications in logistics and SCM is receiving significant attention in the academic world. Moreover, the data presented in the table reveal a collaboration index of 3.36. This metric quantifies the extent of collaborative efforts among researchers in a particular field. In this case, it represents the average number of authors per article in the literature related to drone applications in logistics and SCM. The collaboration index can be interpreted as researchers working together in small- to medium-sized communities to address various aspects of drones.

Figure 3 shows the annual number of publications related to drone applications in logistics and SCM from 2009 to 2023. It is essential to note that the data for 2023 are incomplete, as it only includes journal articles published up to May 2023. Despite this, the annual growth rate for publications in this domain is calculated at an impressive 29.24%. This rapid growth rate demonstrates a burgeoning interest and an

Table 3 Main bibliometric data related to the selected sample of journal articles

Description	Results
Main information about data	
Timespan	2009:2023
Sources (Journals)	57
Documents	96
Average years from publication	2.77
Average citations per documents	27.39
Average citations per year per doc	5.993
References	5933
Document types	
Article	92
Review	4
Document contents	
Keywords Plus (ID)	578
Author's Keywords (DE)	389
AUTHORS	
Authors	306
Author Appearances	328
Authors of single-authored documents	7
Authors of multi-authored documents	299
Authors collaboration	
Single-authored documents	7
Documents per Author	0.314
Authors per Document	3.19
Co-Authors per Documents	3.42
Collaboration Index	3.36

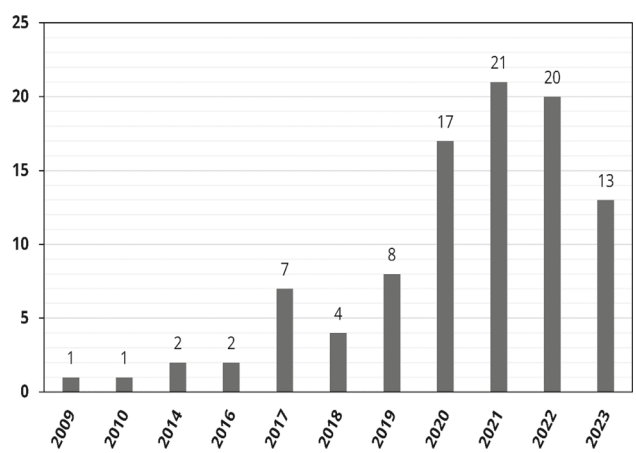


Fig. 3 Annual number of selected articles

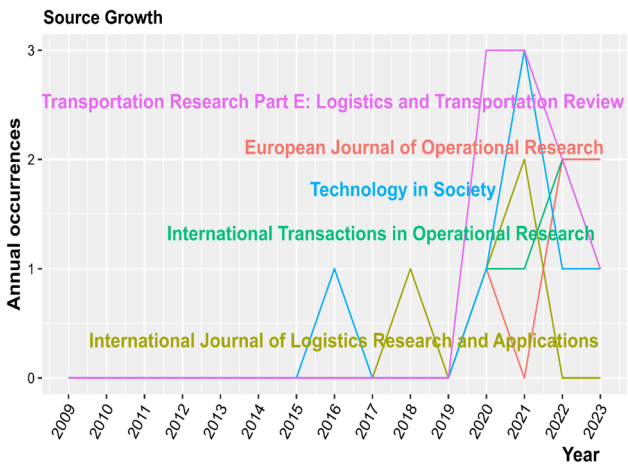


Fig. 4 Most prolific journals

expanding body of research in the field of drone applications in logistics and SCM. This increasing academic attention reflects the recognition of the potential impact and value of drone technology in revolutionizing the logistics and SCM sectors.

Figure 4 shows the five most productive academic journals in the field of drone applications in logistics and SCM. These journals have published the highest number of articles on the topic, indicating their significant contribution to the research landscape in this domain. *Transportation Research Part E: Logistics and Transportation Review* represents the most productive journal, publishing nine articles. This journal specifically focuses on transportation and logistics, making it an ideal platform for disseminating research on drone technology in logistics and SCM. The second most productive, *Technology in Society*, published seven articles on the subject. This journal's broader scope suggests that research on drone applications in logistics and SCM is not only limited to technical aspects but also encompasses social, ethical, and policy-related considerations. *The European Journal of Operational Research* and *International Transactions in Operational Research* are tied in third place, with each journal publishing five articles on drone applications in logistics and SCM. Both journals specialize in operational research, emphasizing the importance of analytical and optimization techniques in understanding and enhancing the role of drone technology in these sectors. Finally, the *International Journal of Logistics Research and Applications* contributes four articles and showcases its commitment to exploring innovative solutions and applications for improving logistics processes, including the integration of drone technology.

The fundamental goal of this research is to develop an LDA model. The researchers used ten keywords and their associated weights to analyze the model's recommended ten topics. An inductive analysis was performed to determine what topics were connected to the retrieved keywords. Ten topics emerge from the analysis, which represents a cross section of current research interests and foci in the field of drone technology in logistics and SCM. Topic 2, focusing on drones for humanitarian logistics and SCM, and Topic 3, centring on the optimization of last-mile delivery using drones and trucks in urban logistics, emerged as the most discussed themes within the literature. The prominence of Topic 2 highlights the increasing recognition of the potential benefits that drone technology can bring to humanitarian efforts [28, 29]. Rejeb et al. [7] note that drones can play a crucial role in delivering essential supplies, medicines, and other resources to disaster-stricken or hard-to-reach areas, particularly in situations where traditional means of transportation are insufficient or unsafe. The significant attention devoted to this topic emphasizes the importance of exploring innovative solutions that leverage drone technology to enhance the efficiency, responsiveness, and effectiveness of humanitarian logistics and SCM. Furthermore, the centrality of Topic 3 in the literature reflects the growing interest in developing more efficient and sustainable delivery solutions within the logistics industry. As urban populations continue to grow and e-commerce demand increases, last-mile delivery becomes a critical challenge for logistics and SCM [30]. As a result, the use of drones in conjunction with traditional delivery vehicles, such as trucks, has the potential to optimize last-mile delivery by reducing delivery times, decreasing operational costs, and minimizing environmental impacts. With PyLDAvis, a Python package created by Sievert and Shirley [31], it is simple to grasp the importance of the weights of the selected topics in the LDA model. Each featured topic is shown as a corresponding colored circle on a 2D map (Fig. 5) generated using PyLDAvis. Topic 2 has the largest circle size, suggesting the most influence. Moreover, the intertopical distance map displays an overlap between Topic 3 (Optimization of last-mile delivery using drones and trucks in urban logistics) and Topic 5 (Drone delivery route optimization and logistical problem-solving). This overlap suggests that there is a strong interconnection between these two themes in the literature because they both deal with aspects of drone utilization in logistics, specifically concerning delivery optimization. The overlapping nature of these two topics also indicates that researchers are exploring various dimensions of delivery optimization with drones, covering both broader issues, such as integrating drones with traditional delivery vehicles like trucks (Topic 3), and more focused aspects, such as route optimization and logistical problem-solving.

Besides visualizing the topics, the pyLDAvis tool provides a list of important keywords used to define the topics. For this study, the top 30 most significant keywords were identified and analyzed, with a focus on understanding the relationship between them. Figure 6 presents the findings and

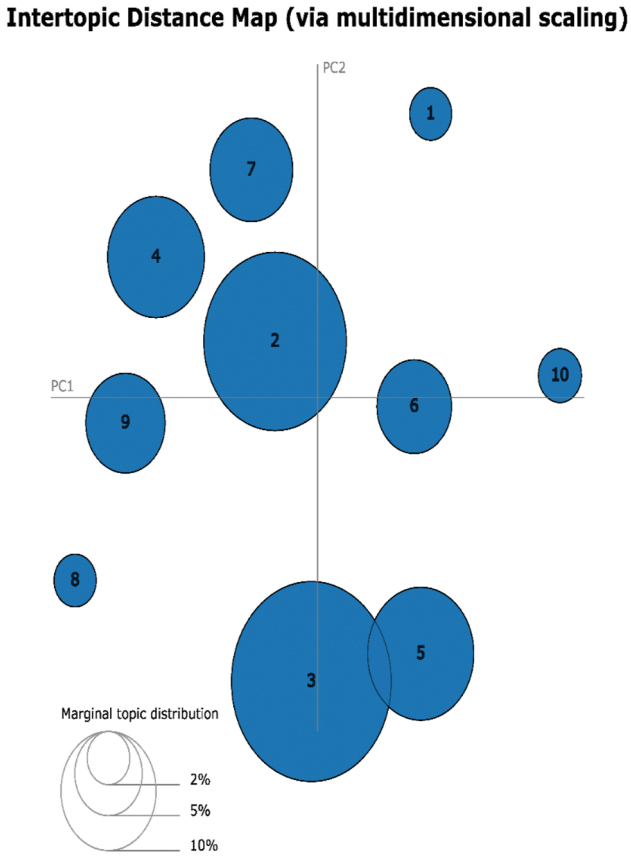


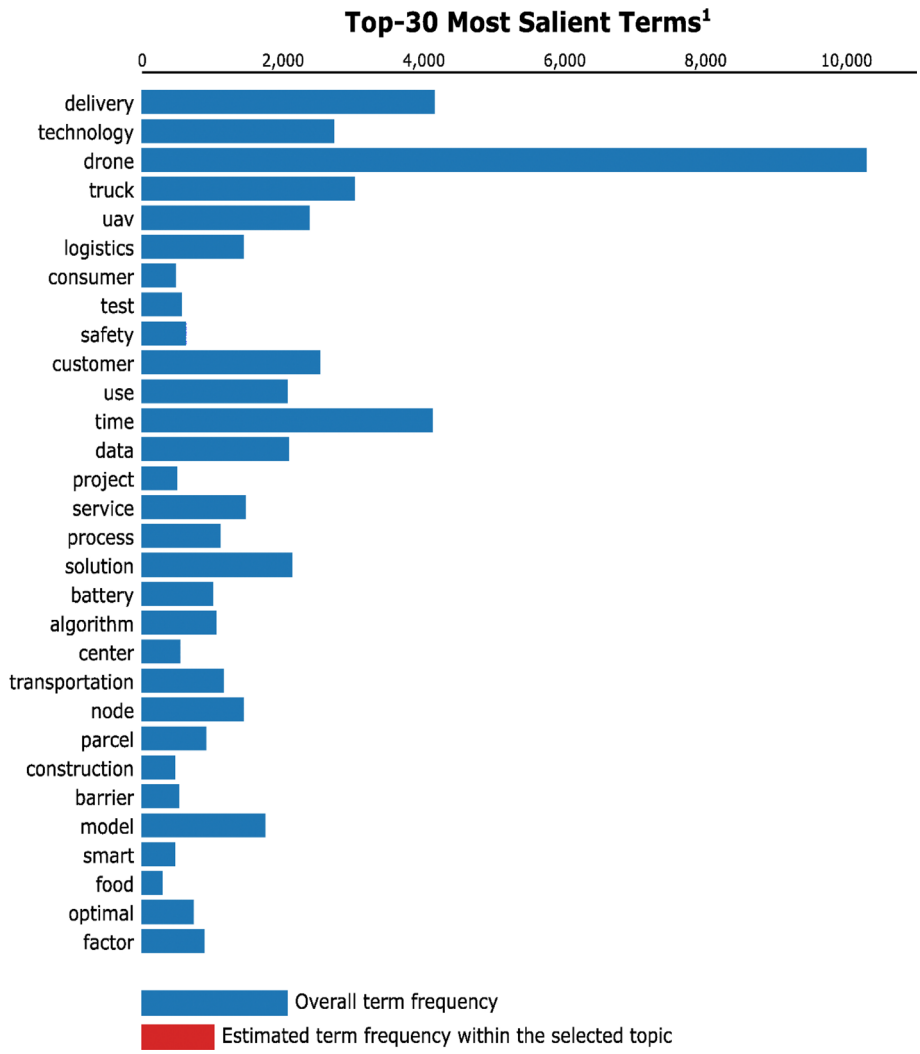
Fig. 5 Intertopical distance map

highlights the prominence of the keywords "drone," "delivery," "time," "truck," and "customer." The high frequency of the keyword "drone" is expected since it represents the primary technology under investigation. The prominence of the keyword "delivery" demonstrates the focus on exploring how drones can improve the efficiency and effectiveness of delivering products and services in logistics and SCM [6]. In this regard, scholars are particularly interested in examining how the technology can reduce delivery times, increase responsiveness, and optimize scheduling in logistics and SCM processes [32–34]. This focus on time suggests that time-sensitive aspects of logistics and SCM, such as last-mile delivery or time-critical shipments, are significant areas of research. The keyword "truck" reflects the integration of drones with trucks to achieve better overall efficiency and sustainability in delivery processes [35, 36]. Finally, the keyword "customer" highlights the importance of understanding the implications of drone technology for end-users in logistics and SCM. Scholars recognize that customer satisfaction and preferences are critical factors that need to be considered when evaluating the potential benefits and challenges of implementing drones [30].

The findings of the model were validated by using a code to analyze the topic distribution in the selected journal articles and to examine the impact of each topic on those articles. According to the data, there are at least five articles that touch on each of the identified topics. The dominance of the second topic, which examines drone applications for humanitarian logistics and SCM, highlights the broader societal implications of the technology, going beyond more commercial or economic concerns to address pressing humanitarian challenges. To a lesser extent, Topic 3 is discussed in 48 articles. This topic underscores the urgency of finding innovative solutions to improve the efficiency and sustainability of urban logistics systems, which are becoming strained due to rapid urbanization [37] and the proliferation of e-commerce activities [38]. Topic 4, which is discussed in 25 articles and focuses on the barriers and factors affecting the use of drones in logistics and humanitarian operations, highlights the importance of understanding the challenges and constraints associated with drone technology implementation. The attention given to this topic reflects a growing recognition among scholars that, despite the significant potential of drones, there are various practical, regulatory, and social issues that need to be overcome. These may include factors such as safety concerns, privacy issues, legal and regulatory frameworks, technological limitations, and societal acceptance [7]. Unlike Topics 2, 3, and 4, Topic 1 (Drones safety and utility) and Topic 8 (Drone Delivery and Consumer Behavior in Food Logistics) are the least studied among the identified topics (Table 4). One plausible explanation for this discrepancy could be attributed to the nascent stage of development and integration of drone technology in these specific domains (Fig. 7).

An analysis of the distribution of journals according to the ten topics reveals that a few topics have a substantial number of publications, whereas several others remain underrepresented. Table 5 outlines the distribution, with a minimum threshold of four articles per journal. It is evident that Topics 2, 3, 5, and 9 have garnered considerable attention from scholars, with multiple journals publishing articles on these themes. For instance, Topic 1 has been extensively covered in journals like *Technology in Society* and the *International Journal of Logistics Research and Applications*, while Topic 3 has been the focus of research in *Transportation Research Part E: Logistics and Transportation Review* and the *International Transactions in Operational Research*. Similarly, Topics 5 and 9 have attracted the attention from outlets such as the *European Journal of Operational Research* and *Technology in Society*, respectively. This concentration of research interest may be indicative of the relevance or potential for innovation in these topics. For instance, optimization of last-mile delivery, drone delivery route optimization, and public acceptance of drone technology are deemed crucial for the successful integration of drones into logistics and SCM.

Fig. 6 Top 30 most salient terms



In contrast, Topics 1, 4, 6, 7, 8, and 10 do not have any journals meeting the minimum threshold of four articles, signaling a dearth of research activities in these topics. This underrepresentation may be attributed to several factors, including the immaturity of drone technology in these areas or the complexities associated with these topics. It is also possible that research priorities of academic institutions have contributed to this disparity, with resources and support being directed toward the more established or widely recognized research areas.

4 Topics Discussion

4.1 Drones Safety and Utility

Upon examining Topic 1, which has been labeled as "Drones safety and utility," a variety of terms emerge that contribute to a deeper understanding of the research focus within this cluster. According to Table 4, these terms encompass diverse aspects, including safety, criterion, ranking, structure, utility, mission, and design, among others. A comprehensive analysis and interpretation of the topic, considering the interplay of these terms, can shed light on these underlying themes and potential research directions in this area. At the core of Topic 1, the emphasis on safety and utility highlights the primary concerns of researchers, policymakers, and practitioners in the domain of drone applications [39]. As such, Zhang et al. [40] argue that the safe operation of drones in various contexts, such as firefighting, monitoring, and community use, is of paramount importance to mitigate potential risks and hazards to both human lives and property. Moreover, the utility of drones is a critical aspect that warrants investigation since it pertains to the efficacy, adaptability, and suitability of drone technology in addressing the diverse requirements of multiple sectors and missions [41]. The presence of terms such as criterion, ranking, and structure signifies the importance of developing systematic frameworks and methodologies to

Table 4 LDA model results

Topic	Keywords	Topics
1	0.017*"safety" + 0.011*"drone" + 0.010*"criterion" + 0.009*"ranking" + 0.008*"structure" + 0.008*"utility" + 0.006*"work" + 0.006*"level" + 0.006*"case" + 0.006*"high" + 0.005*"mission" + 0.005*"use" + 0.005*"weight" + 0.005*"planning" + 0.005*"uas"	Drones safety and utility
2	0.024*"drone" + 0.013*"data" + 0.012*"technology" + 0.007*"use" + 0.007*"logistics" + 0.006*"disaster" + 0.006*"system" + 0.006*"transportation" + 0.006*"application" + 0.005*"management" + 0.004*"case" + 0.004*"operation" + 0.004*"big" + 0.004*"area" + 0.004*"humanitarian"	Drones for humanitarian logistics and SCM
3	0.033*"drone" + 0.022*"truck" + 0.018*"delivery" + 0.017*"time" + 0.016*"customer" + 0.011*"cost" + 0.010*"solution" + 0.008*"problem" + 0.007*"node" + 0.006*"number" + 0.006*"instance" + 0.006*"vehicle" + 0.006*"location" + 0.006*"total" + 0.006*"operation"	Optimization of last-mile delivery using drones and trucks in urban logistics
4	0.019*"uav" + 0.015*"drone" + 0.011*"delivery" + 0.010*"barrier" + 0.008*"factor" + 0.006*"logistics" + 0.006*"privacy" + 0.006*"use" + 0.005*"humanitarian" + 0.004*"technology" + 0.004*"system" + 0.004*"country" + 0.004*"vehicle" + 0.004*"company" + 0.004*"autonomous"	Barriers and factors affecting the use of drones in logistics and humanitarian operations
5	0.014*"time" + 0.012*"drone" + 0.010*"problem" + 0.010*"battery" + 0.009*"solution" + 0.008*"node" + 0.007*"value" + 0.006*"model" + 0.006*"number" + 0.006*"algorithm" + 0.006*"optimal" + 0.006*"instance" + 0.006*"set" + 0.005*"customer" + 0.004*"heuristic"	Drone delivery route optimization and logistical problem-solving
6	0.025*"drone" + 0.015*"project" + 0.014*"process" + 0.012*"construction" + 0.011*"time" + 0.010*"audit" + 0.009*"flight" + 0.007*"asset" + 0.007*"use" + 0.006*"technology" + 0.006*"design" + 0.005*"uas" + 0.005*"inventory" + 0.005*"count" + 0.005*"production"	Drones in construction projects and asset management
7	0.020*"technology" + 0.010*"system" + 0.009*"data" + 0.007*"uav" + 0.007*"enabler" + 0.007*"adoption" + 0.006*"drone" + 0.006*"application" + 0.006*"smart" + 0.005*"supply chain" + 0.005*"city" + 0.004*"challenge" + 0.004*"table" + 0.004*"management" + 0.004*"integration"	Technology adoption and impact on supply chain and city operations
8	0.032*"consumer" + 0.025*"technology" + 0.019*"delivery" + 0.018*"drone" + 0.017*"intention" + 0.015*"food" + 0.013*"logistics" + 0.009*"habit" + 0.009*"behavioral" + 0.009*"usage" + 0.009*"service" + 0.008*"attitude" + 0.007*"use" + 0.006*"model" + 0.006*"behavior"	Drone delivery and consumer behavior in food logistics
9	0.032*"drone" + 0.019*"delivery" + 0.010*"belief" + 0.008*"risk" + 0.008*"use" + 0.007*"service" + 0.006*"concern" + 0.006*"parcel" + 0.006*"job" + 0.006*"time" + 0.006*"public" + 0.005*"transportation" + 0.004*"group" + 0.004*"participant" + 0.004*"people"	Public acceptance and concerns regarding drone technology in transportation and logistics
10	0.022*"test" + 0.020*"swab" + 0.017*"barcode" + 0.012*"laboratory" + 0.012*"processing" + 0.009*"reagent" + 0.009*"uav" + 0.008*"landing" + 0.007*"frame" + 0.007*"center" + 0.006*"type" + 0.006*"transportation" + 0.006*"kit" + 0.006*"region" + 0.005*"time"	Drones in medical/laboratory transport

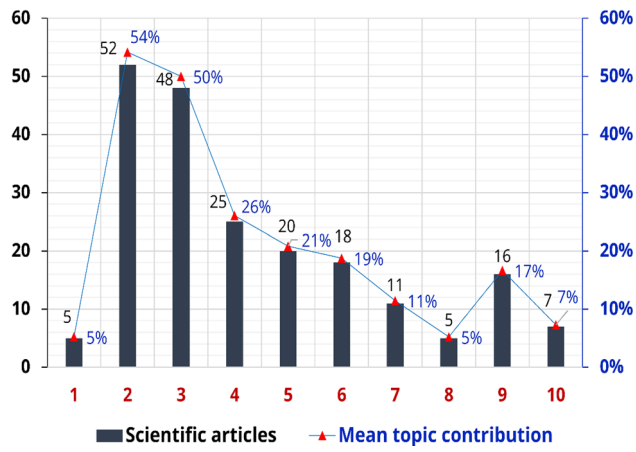


Fig. 7 Distribution of topics among selected articles

evaluate and compare different drone designs and applications based on safety and utility. These methodologies may involve the use of advanced decision-making tools such as fuzzy analytic hierarchy process (fuzzy AHP) and DEMATEL, which can help stakeholders navigate the complexities of the drone landscape and make informed choices regarding the selection and implementation of drone technologies [40]. Additionally, terms such as mission, planning, and design underscore the need to consider the context in which drones operate, as well as the specific objectives they aim to achieve. In this regard, Rodrigues Santos de Melo and Bastos Costa [39] state that there is a need for a comprehensive approach to designing drone systems that are not only safe and useful but also tailored to the unique demands of various scenarios, including firefighting, personal use, or building and infrastructure monitoring. Overall, the analysis of Topic 1 reveals a multifaceted research area that highlights the urgency to ensure the safe and effective integration of drone technology into diverse applications and contexts.

4.2 Drones for Humanitarian Logistics and SCM

Topic 2, labeled as "Drones for humanitarian logistics and SCM," contains a range of terms that collectively provide insights into the applications of drones for disaster relief operations. These terms include drone, data, technology, logistics, disaster, system, and transportation, among others. The focus of the topic is on the growing interest in leveraging drone technology for humanitarian purposes, particularly in the context of humanitarian SCM [7]. More specifically, Jeong et al. [42] indicate that this interest stems from the recognition of the potential of drones to revolutionize the transportation and delivery of essential products and services during disasters, emergencies, or in areas with limited infrastructure. Tatham [28] argues that the emphasis on the application of drone technology in humanitarian logistics showcases the transformative power of this technology in overcoming the challenges faced by disaster-stricken communities and aid organizations. Furthermore, terms like data, system, and information allude to the importance of harnessing the vast amounts of data generated and processed by drones in order to optimize their deployment and effectiveness in humanitarian logistics and SCM. According to Akhtar et al. [43], this data-driven approach entails the integration of advanced analytics, big data techniques, and information management, which can facilitate real-time decision-making, improve operational efficiency, and enhance the overall responsiveness of humanitarian supply chains. Furthermore, the presence of the terms disaster, medical, warehouse, and delivery underscores the diverse range of applications and scenarios in which drones can be employed to support humanitarian logistics and SCM. These applications may include but are not limited to disaster relief operations [1], medical supply deliveries [44], and remote warehousing and inventory management [38]. In summary, the versatility and adaptability of drone technology make it a valuable asset in addressing the unique challenges and constraints faced by humanitarian organizations.

Table 5 Distribution of journals based on the identified topics (threshold = 4 articles per journal)

Topic	Source
1	/
2	- Technology in Society (4) International Journal of Logistics Research and Applications (4)
3	- Transportation Research Part E: Logistics and Transportation Review (8) International Transactions in Operational Research (5)
4	/
5	- Transportation Research Part E: Logistics and Transportation Review (5) - European Journal of Operational Research (4)
6	/
7	/
8	/
9	- Technology in Society (4)
10	/

4.3 Optimization of Last-Mile Delivery Using Drones and Trucks in Urban Logistics

Based on the findings of the LDA model, Topic 3 contains various terms related to the optimization of last-mile delivery using drones and trucks in urban logistics. These

include drone, truck, delivery, time, customer, cost, solution, and model, among others. Several scholars highlight the role of combining drones and trucks in overcoming the challenges associated with urban logistics. For example, Lemardelé et al. [3] explore the optimal pairing of drones and ground-autonomous delivery devices with the features of urban environments to streamline last-mile logistics operations. The findings indicate that denser neighborhoods are more economically suited for ground autonomous delivery devices, while truck-launched delivery drones are better suited for less dense and larger service regions. Rave el al. [45] suggest a planning model for logistics service providers that reduces delivery costs in rural areas by integrating aerial drones and trucks and conclude that using both drones launched from trucks and dedicated drone stations leads to the greatest cost-savings. Amorosi et al. [46] present a model that utilizes a multi-visit drone with limited endurance and a base vehicle that moves freely to visit multiple targets of varying shapes using a mixed integer second-order cone program and a refined metaheuristic to minimize the overall weight distance traveled by both vehicles. The findings of the authors confirm the versatility and applicability of drones in various scenarios. Vu et al. [47] investigate the two-echelon vehicle routing problem that employs a truck and a group of drones to reduce the completion time instead of transportation costs. Finally, Salama and Srinivas [48] propose a new approach for last-mile package deliveries using a truck and multiple drones that can launch and recover at flexible sites and develop a model to optimize vehicle assignment, routing, and scheduling. The results show significant improvement in delivery efficiency and provide insights into drone utilization and site selection. Overall, Topic 3 aims to contribute to the advancement of drone and truck integration in last-mile delivery operations and promote more efficient, cost-effective, and customer-centric urban logistics solutions.

4.4 Barriers and Factors Affecting the Use of Drones in Logistics and Humanitarian Operations

Topic 4 is labeled as "Barriers and factors affecting the use of drones in logistics and humanitarian operations," encompassing a range of terms that collectively provide a nuanced perspective into this complex research area. Several scholars contribute valuable insights to Topic 4. For example, Kamat et al. [49] examine the obstacles that prevent the implementation of drones in Indian humanitarian logistics by using a two-step approach involving the grey-DEMATEL and analytic network process (ANP). Their study identifies expensive commercial solutions and high transport energy costs as major factors that hinder the adoption of drones in the humanitarian logistics industry, causing delays in implementation. Similarly, Kamat et al. [50] study the barriers to implementing drones in humanitarian logistics and propose an approach to calculate a drone implementation hindrance index (DIHI). The findings indicate that inadequate government regulations and lack of technological awareness are major hindrances in both developed and developing countries, providing policymakers and practitioners with insights to facilitate drone implementation in their respective countries' humanitarian sectors. In the context of business logistics, Sah et al. [51] attempt to prioritize the barriers hindering the adoption of drones by using the fuzzy Delphi method and the analytic hierarchy process. Regulations and privacy and security threats are identified as the most significant barriers, followed by public perception and psychological, environmental, technical issues, and economic aspects. Therefore, the successful integration of drones in logistics and SCM is not without challenges since the use of drones is influenced by a multitude of factors, which can be technological, regulatory, societal, and economic in nature. For instance, the technical barriers may include limitations in drone capabilities, such as payload capacity, battery life, and navigational accuracy. Regulatory barriers, on the other hand, may encompass laws and regulations related to airspace use, privacy concerns, and safety standards. Societal barriers can involve public acceptance and perception of drone technology, while economic factors may include the cost-effectiveness of drone operations compared to traditional delivery methods. Consequently, there is a need for a comprehensive approach to understanding and addressing the barriers and factors influencing drone adoption in logistics and SCM.

4.5 Drone Delivery Route Optimization and Logistical Problem-Solving

Topic 5, labeled "Drone delivery route optimization and logistical problem-solving," is a fascinating area of drone application studies. This topic delves into the mathematical intricacies and algorithmic complexities associated with optimizing drone delivery routes, manifesting the intersection of operations research and SCM. For example, Zhang et al. [29] propose a drone arc routing problem to assess a humanitarian transportation network after a disaster, with the goal of maximizing the collected arc informative profits within a predefined time limit. Their study finds that the proposed graph information technique and efficient exact branch-and-price framework can solve the problem efficiently and achieve a 30% improvement in the objective value compared to the traditional arc orienteering problem. Asadi et al. [32] focus on mitigating the long recharge times and battery degradation of drones and electric vehicles through the use of swap stations and propose a stochastic scheduling problem for determining the charging, discharging, and replacement decisions at these stations over time. Baniasadi et al. [52] examine the difficulties in solving the clustered generalized traveling salesman (CGTPS), an extension of the classical

traveling salesman problem (TSP), by proposing an efficient transformation from CGTSP to TSP. The approach of the authors can be applied to modern logistics problems, including drone-assisted parcel delivery services. The optimization of logistics operations through the use of drones in combination with traditional modes of transportation, such as trucks and courier services, has also been highlighted in the literature. For instance, Pei et al. [53] investigate revenue management in a shared drone delivery system by formulating a queueing model that takes into account customer behavior, pricing for courier service, and maintained fleet size. The results indicate that the proposed approximation methods help stabilize the system's performance and only require a slight change in price to reach optimal conditions. Mbiadou Saleu et al.[33] extend the parallel drone scheduling TPS to a parallel drone scheduling multiple TPS and propose a hybrid metaheuristic solution approach. The suggested approach is validated through an experimental campaign and provides insights into the drone delivery system. Finally, Carlsson and Song [54] evaluate the efficiency of a delivery system that uses a drone to provide service to customers while making return trips to a truck, and find that the efficiency gain is square-root-proportional to the difference in travel speeds between the truck and the drone. In summary, Topic 5 represents a rich vein of research that employs sophisticated mathematical and computational methodologies to enhance the efficiency and effectiveness of drone deliveries. This topic not only contributes to the theoretical development of the drone field but also has significant practical implications as it can help logistics companies to enhance their delivery drone operations and provide better service to their customers.

4.6 Drones in Construction Projects and Asset Management

Topic 6 revolves around the application of drones in managing, monitoring, and optimizing construction projects. Notable keywords such as "project," "construction," "process," and "time" suggest a focus on improving project management and operational efficiency in construction projects. Related to this topic, Irizarry and Costa [55] analyze potential uses of visual assets obtained from drones in construction management tasks through an exploratory case study. The findings of the authors highlight the importance of drones in monitoring project progress, handling logistics on the job site, assessing safety conditions, and conducting quality inspections. Rodrigues Santos de Melo & Bastos Costa [39] present a conceptual framework and a safety monitoring protocol utilizing drones to improve safety management and lowering accident rates in the construction industry, highlighting the role that resilience engineering and drones play in supporting the safety planning and control process. Likewise, Li and Liu [56] explore the current uses of multirotor drones in the construction sector, evaluate their advantages, and examine their future potential. The findings show that drones improve worker safety, increase cost-effectiveness, and reduce carbon emissions. In addition to these benefits, Chen et al. [57] demonstrate that drones can contribute to optimal work efficiency, increased productivity, and high quality, particularly through their ability to acquire data. Furthermore, the potential use of drone technologies to enhance audit quality and efficiency, both in terms of inventory counts and examination and inspection applications, has been discussed in a few studies. For example, Appelbaum and Nehmer [58] examine the feasibility of adopting drones in audit environments and provide a tailored architecture for prototyping inventory counts in expansive warehouses and outdoor stockrooms. Christ et al. [59] determine whether or not adopting drones and automated counting software can improve audit quality. Their results suggest that auditors can use these technologies to complete inventory counts more quickly and effectively while delivering higher-quality audit documentation. Finally, Ali et al. [38] explore the potential drone applications in smart warehouse management and conclude that the technology can optimize inventory management, intro-logistics, inspections, and surveillance in warehouses. In conclusion, Topic 6 represents a comprehensive investigation into the use of drones within the construction industry and asset management. This research topic is significant for advancing technological integration in construction project management and shaping future industry practices regarding safety, quality control, and asset management.

4.7 Technology Adoption and Impact on Supply Chain and City Operations

Labeled as "Technology adoption and impact on supply chain and city operations," Topic 7 encompasses the evaluation of the role and impact of drones within the larger landscape of urban operations and SCM. The coupling of "supply chain" and "city" indicates an exploration of the implications of drone integration at both the organizational and urban levels. This could include studies into how drones influence logistics and SCM operations and how they can be leveraged to enhance smart city operations. Representative studies addressing Topic 7 include Ali et al. [60], who create a framework for ranking the determinants of drone deployment and enablers for adopting the technology using both expert feedback and quantitative approaches. Similarly, Rathore et al. [61] identify the main barriers that hamper the effective integration of digital technologies, including drones, in the logistics sector and explore their interrelationships. The authors find that lack of top management

support represents a critical barrier, while legal and regulatory frameworks, infrastructure, and reluctance to change are driving barriers. In the quest to redefine urban landscapes, several researchers explore how drones can be seamlessly integrated into city operations and transport services. As a case in point, Mohamed et al. [62] summarize the applications of drones in smart cities, their implications, and the technical and non-technical challenges facing their integration. According to the authors, drones can optimize resource use, improve city operations, and increase citizen engagement; meanwhile, safety, privacy, and ethical issues need to be addressed to ensure the success of the technology in smart cities. In another study, Bubelíny et al. [63] argue that drones can be used for traffic monitoring and management because the technology provides comprehensive coverage of entire intersections, parking areas, and nodes to collect data on vehicle volume and congestion. This information can include the number of vehicles passing through specific points and areas, as well as the identification of heavily trafficked roads. Therefore, the studies encapsulated within Topic 7 provide valuable insights into the promising yet complex interplay between drones, SCM, and urban operations. These research efforts underscore the necessity of a multifaceted approach to drone adoption, one that balances the operational benefits and transformative potential of the technology with the necessary attention to prevailing barriers such as regulatory frameworks, infrastructure requirements, and societal acceptance. As the march of progress continues, it is crucial that academics, industry, and policymakers continue to collaborate, ensuring the successful deployment of drones in our cities and supply chains while addressing the incumbent challenges head-on.

4.8 Drone Delivery and Consumer Behavior in Food Logistics

Topic 8 is denominated as "Drone delivery and consumer behavior in food logistics," and it delves into the intricate dynamics of consumer acceptance, usage, and overall behavioral responses to the introduction and implementation of drone delivery in the food sector. The discourse on this topic largely revolves around factors influencing consumers' intention to use drone services, their attitude toward this novel mode of delivery, and their behavioral habits in relation to technology-driven food logistics. For example, Khalil et al. [64] explore the impact of functional and psychological barriers on consumers' aversion to food delivery services relying on drones. The authors find that risk, usage, experience, and image barriers play a key role in this reluctance. Pinto et al. [65] suggest an optimization model to identify the optimal placement of charging stations for a fleet of drones carrying food products considering the technological and legal barriers to technology adoption in food logistics. Akhtar et al. [43] argue that drones can be used to provide food supplies where logistical challenges exist. The use of drones in food logistics can improve service quality, which is an essential performance indicator for humanitarian excellence and timely response. Moreover, Bafouni-Kotta et al. [66] report the willingness of respondents to pay for drone delivery services in the food category. This suggests that consumers see value in the speed and efficiency of drone delivery for food as it may be faster and more reliable than traditional delivery methods, especially in densely populated areas or areas with challenging terrain. Additionally, the use of drones for delivery can offer greater flexibility and convenience for consumers, who can order food from a wide range of restaurants or shops and receive it quickly and efficiently [6]. Furthermore, Topic 8 embraces the utilization of behavioral theories and models to elucidate the determinants of consumers' acceptance and use of drone delivery. For instance, constructs from established theories such as the Unified Theory of Acceptance and Use of Technology (UTAUT) have been employed to conceptualize and measure factors like performance expectancy, effort expectancy, and social influence that could impact consumer adoption [67]. In conclusion, Topic 8 provides a comprehensive examination of the intersection between drone delivery technology, consumer behavior, and food logistics. The scholarship in this area not only maps out the influences shaping consumer responses, but it also pinpoints strategies that could ameliorate barriers and optimize benefits in drone-enabled food delivery. Moreover, the use of behavioral theories underpins the rigorous, multidimensional approach taken to understand this complex ecosystem. Ultimately, as we navigate an era of accelerated technological disruption and increased consumer empowerment, the insights gleaned from these studies are instrumental in driving innovation, enhancing service quality, and championing consumer-centric approaches in the realm of food logistics.

4.9 Public Acceptance and Concerns Regarding Drone Technology in Transportation and Logistics

Under the umbrella of "Public acceptance and concerns regarding drone technology in transportation and logistics," Topic 9 ventures into the complex interplay of public attitudes, apprehensions, and adaptability in relation to the deployment of drones within transportation and logistics sectors. This topic serves as a platform for scholars to unravel the intricacies of public perceptions and the diverse factors that govern the acceptance of drones for delivery and transportation purposes. For example, Merkert et al. [68] examine consumer preferences for last-mile parcel delivery options, including drones, and find that consumers generally favor traditional postal delivery over drone delivery, but drones

could gain a significant market share if they can provide faster and more affordable delivery. However, both postal delivery options and drones have become less desirable compared to parcel lockers when there is no secure location to leave a package at a recipient's address. Furthermore, Zhu et al. [69] analyze the structure of the public's risk beliefs toward drone delivery and its impact on risk communication and management. The authors further use network analysis to identify key risk beliefs and find that messages targeting structurally central beliefs are more effective in mitigating risk perceptions, offering useful insights into risk perception related to drone delivery and how to improve risk communication. Kellermann and Fischer [70] explore how the public perceives the use of transport drones for parcel and passenger transport in urban areas. Based on qualitative results from focus group discussions in Germany, the study shows that participants held varied views on the topic, with a focus on object, subject, and context-related factors. Finally, Smith et al. [71] analyze the existing research on public acceptance of drones in transport systems, pointing out the ambiguity in the concerns due to the combination of various drone applications. The authors emphasize the need for building practical familiarity and understanding of drone use to enable informed debates among policymakers and stakeholders. From the methodological perspective, Topic 9 seems to be grounded in a participatory approach, often involving direct engagement with the public, for example, through surveys, focus groups, or choice modeling.

4.10 Drones in Medical/Laboratory Transport

Navigating the nexus of drones and healthcare logistics, Topic 10—"Drones in medical/laboratory transport"—unfolds as an exploration of the promising potential of drone technology in medical supply chains. The topic exhibits a progressive inquiry into the role of drones in streamlining medical supply chains, enhancing the speed and efficiency of laboratory sample transportation, and providing solutions for healthcare logistics in geographically challenging regions. In this regard, Colajanni et al. [72] develop an optimization model for the supply chain involved in swab tests during the COVID-19 pandemic, which incorporates the sharing of reagents and the use of drones with 5G technology. According to the results of the numerical simulations, drones have been demonstrated effective in transporting swabs, kits, and reagents. In addition, Koshta et al. [73] note drones can enhance the efficiency and effectiveness of supply chains for humanitarian aid to combat COVID-19, particularly in food, administrative, and healthcare supply chains. In the African context, Nyaaba and Ayamga [74] argue that drones can facilitate the delivery of blood, automated external defibrillators (AEDs), drugs, vaccines, and laboratory test samples. However, it is essential to adopt inclusive innovation strategies, such as promoting public awareness of the technology, providing training for health personnel on dispatching medical supplies through drones, and working with regulators in order to maintain the utilization of medical drones in Africa and ensure use safety. Finally, Pulver and Wei [75] propose a new spatial optimization model to aid in the implementation of a network of medical drones equipped with on-board AEDs to decrease response times for out-of-hospital cardiac arrests by strategically placing drones and launch sites, thereby enhancing backup coverage with minimal loss of primary coverage. Consequently, this discourse clearly demonstrates the compelling intersection of drone technology and healthcare logistics. As the implications of such integration unfold, it becomes increasingly evident that drones hold the potential to revolutionize medical supply chains and emergency response systems, thereby enhancing healthcare outcomes. Hence, the exploration of drones in medical and laboratory transport within Topic 10 offers invaluable insights for both the technological and healthcare sectors, shedding light on the transformative power of this emerging technology when applied to critical areas of public health.

5 Conclusions

5.1 Summary

In the ever-evolving world of logistics and SCM, organizations face increasing complexities as they navigate the challenges of global markets, customer demands, and regulatory compliance. The role of new technologies in mitigating these challenges and enhancing logistics and SCM processes cannot be understated. Emerging technologies such as artificial intelligence, IoT, and advanced analytics are instrumental in providing more efficient, cost-effective, and resilient solutions for modern SCM. Among these innovative technologies, drones have emerged as a disruptive force with the potential to transform logistics and SCM applications. From last-mile delivery and inventory management to facilitating emergency response and humanitarian aid, drones offer unprecedented capabilities in terms of flexibility, speed, and cost-savings. The technology has the power to revolutionize the way goods are transported, particularly in remote and hard-to-reach areas, and can significantly streamline supply chain processes.

Despite the growing interest in drone applications in logistics and SCM, there is a noticeable gap in the literature regarding the use of latent Dirichlet allocation (LDA)-based topic modeling to systematically analyze the research landscape at the intersection of drones, logistics, and SCM. As a result, a thorough examination using LDA-based topic modeling would provide valuable insights into the current state of research and identify key areas of focus and potential avenues

for future studies. Such analysis could contribute to a deeper understanding of the implications of drones in logistics and SCM and guide both academia and industry in harnessing the transformative power of drones to revolutionize the field.

Based on a sample of 96 journal articles, this study reveals an escalating academic interest in drone applications in logistics and SCM, underscored by a 29.24% annual growth rate in related publications. The shift from sporadic publications before 2017 to a continuous surge afterward mirrors the maturing of drone technology and its growing applicability. Furthermore, contributions from diverse academic journals illustrate a comprehensive and multidisciplinary exploration of drones in logistics and SCM, encompassing technical, social, ethical, and policy-oriented perspectives. Overall, the results of the LDA model reveal that literature pertaining to drones revolves mainly around ten topics: (1) Drones safety and utility, (2) Drones for humanitarian logistics and SCM, (3) Optimization of last-mile delivery using drones and trucks in urban logistics, (4) Barriers and factors affecting the use of drones in logistics and humanitarian operations, (5) Drone delivery route optimization and logistical problem-solving, (6) Drones in construction projects and asset management, (7) Technology adoption and impact on supply chain and city operations, (8) Drone delivery and consumer behavior in food logistics, (9) Public acceptance and concerns regarding drone technology in transportation and logistics, and (10) Drones in medical/laboratory transport. The concentration of the literature around Topics 2 and 3 is indicative of the pivotal role drones have come to play in these areas. It also reflects an acknowledgment of the pressing to innovate and improve existing systems, especially in humanitarian logistics and last-mile delivery contexts. In terms of humanitarian logistics and SCM, the logistical challenges posed by disaster-stricken and hard-to-reach areas, such as remote regions of conflict zones, have made traditional methods of delivery insufficient. With their ability to overcome terrain and infrastructure obstacles, drones have emerged as an effective solution. Their adoption in this context is garnering considerable attention because drones can expedite delivery times and enhance the efficiency of relief operations, ultimately saving lives and resources. Likewise, the emphasis on optimizing last-mile delivery using drones and trucks in urban logistics highlights the growing demand for fast, efficient, and sustainable delivery solutions in increasingly congested urban environments. Last-mile delivery is often the most costly and time-consuming part of the supply chain. However, the integration of drones into this process can provide an alternative or complementary solution to traditional delivery methods, helping to reduce delivery times, alleviate road congestion, and lower carbon emissions.

The relatively lower prominence of Topics 1 and 8 may be attributed to the current stage of drone technology development and the specific nature of the application areas. For Topic 1, although drone safety is a crucial aspect of drone applications, it may not have emerged as a primary focus of research due to the ongoing advancements in technology that have led to significant improvements in drone safety and reliability. As drone technology becomes more established and widespread, scholars may prioritize the exploration of innovative use cases and applications rather than solely focusing on the technology's safety features. However, it is essential not to overlook the importance of safety and utility because they remain critical for the successful implementation of drones in various sectors. Regarding Topic 8, "Drone delivery and consumer behavior in food logistics," the lesser attention could be attributed to the niche nature of this application area. Food logistics is a specialized domain, and consumer behavior in relation to drone delivery is an even narrower focus. While drone delivery in food logistics presents unique challenges, such as ensuring food quality and maintaining optimal delivery times, it may not have attracted widespread interest due to the technology's nascent stage in this specific context. As drone delivery becomes more common and organizations increasingly explore its applications in food logistics, it is likely that research in this area will expand to better understand the interplay between drone delivery and consumer behavior.

5.2 Future Research Agenda

As we navigate the evolving landscape of drone applications in various sectors, it is imperative to identify the key research questions and directions that can inform and shape future scholarly inquiries. The following research agenda has been structured to encapsulate the most pressing and promising areas of study across each of the ten topics identified in the systematic literature review. This agenda, encapsulated in Table 6, aims to stimulate further academic discourse and inspire innovative research approaches that will contribute to our comprehensive understanding of drone technology's role and potential in transportation, logistics, and supply chain management.

5.3 Research Implications

The systematic review offers several theoretical implications that can guide the development of drone-related scholarship in the fields of transportation, logistics, and supply chain management. First, it provides a comprehensive conceptual map of the current academic landscape concerning drones, offering an integrated perspective that transcends individual disciplines and domains. By synthesizing a wide range of studies, this article delineates key thematic areas and points out theoretical gaps and inconsistencies within the existing literature. Therefore, it offers a more holistic understanding of the complexities and nuances of drone applications.

Table 6 Proposed research agenda for drone applications in logistics and SCM

Topic	Future research questions	Future research directions
Drones safety and utility	How can the safety standards of drones be further improved to mitigate potential risks in various application contexts? How can the utility of drones be maximized across different sectors and mission requirements? What are the key criteria for evaluating the safety and utility of drones in different contexts?	To examine the impact of technological advancements on the safety and utility of drones To explore the development of systematic frameworks for assessing drone safety and utility in various contexts To investigate the role of decision-making tools in evaluating and selecting drone technologies based on safety and utility criteria
Drones for humanitarian logistics and SCM	What are the potential challenges and ethical considerations associated with using drones for humanitarian logistics and SCM? How can drones be used to improve the delivery of medical supplies in disaster-stricken areas? What are the specific design and operational requirements for drones in humanitarian logistics?	To develop drone designs and operational models tailored for humanitarian logistics, taking into account the unique challenges and constraints of disaster scenarios To assess the potential of drone technology in the delivery of medical supplies in disaster-stricken or hard-to-reach areas
Optimization of last-mile delivery using drones and trucks in urban logistics	What are the operational and regulatory challenges associated with drone and truck integration for last-mile delivery in urban areas? How can artificial intelligence and machine learning be applied to optimize drone and truck coordination in urban logistics?	To explore the regulatory landscape and operational challenges that may hinder or promote the integration of drones and trucks in last-mile delivery To propose robust and resilient operational strategies for drone–truck integrated systems that can adapt to varying urban logistics scenarios
Barriers and factors affecting the use of drones in logistics and humanitarian operations	How can the technological limitations of drones be overcome to enhance their applicability in logistics and humanitarian operations? What policy adjustments are needed to address the regulatory challenges associated with drone usage in these sectors?	To innovate to extend drone capabilities, such as payload capacity, battery life, and navigational accuracy, for improved performance in logistics and humanitarian operations To evaluate and propose adjustments to existing policies that hinder drone utilization in logistics and humanitarian operations
Drone delivery route optimization and logistical problem-solving	How can drones and traditional modes of transportation be better integrated to optimize logistical operations? What are the potential benefits and challenges of using drones for logistical problem-solving in different industries and contexts?	To conduct studies to better understand the integration of drones with traditional modes of transportation for optimal logistical operations To investigate the potential benefits and challenges of using drones for logistical problem-solving across various industries and contexts
Drones in construction projects and asset management	How might drones be used to enhance inventory management in construction projects and asset management? What are the impacts of drone technology on the cost-effectiveness and environmental footprint of construction operations?	To investigate further applications of drones to enhance safety measures and quality control in construction projects To explore the potential use of drones in the maintenance, inspection, and lifespan management of constructed assets
Technology adoption and impact on supply chain and city operations	What new methodologies can be developed to effectively integrate drones into urban operations and supply chains? How can drone technology be harnessed to optimize resource usage, enhance city operations, and increase citizen engagement in smart cities?	To develop innovative methodologies to integrate drones effectively into urban operations and supply chains To investigate how drone technology can be utilized to optimize resource usage, improve city operations, and increase citizen engagement in smart cities

Table 6 (continued)

Topic	Future research questions	Future research directions
Drone delivery and consumer behavior in food logistics	How does consumer behavior change with the introduction of drone delivery services in the food industry, and what factors drive these changes? What barriers hinder consumer acceptance of drone-based food delivery, and how can these be overcome?	To conduct comprehensive studies to understand the shifts in consumer behavior with the advent of drone delivery services in the food industry, and identify the key drivers of these changes To identify and explore potential solutions to overcome barriers hindering consumer acceptance of drone-based food delivery
Public acceptance and concerns regarding drone technology in transportation and logistics	How can societal trust in drone technology be cultivated and enhanced, particularly in the context of transportation and logistics? What are the implications of demographic factors like age, gender, education, and occupation on the public acceptance of drone technology in transportation and logistics?	To investigate the strategies to build societal trust in drone technology, specifically in the context of transportation and logistics. This could include the development and testing of interventions aimed at increasing transparency and accountability in drone operations To conduct demographic-specific research to understand how factors such as age, gender, education, and occupation influence public acceptance and concerns regarding drone technology in transportation and logistics
Drones in medical/laboratory transport	How can drone technology improve supply chain efficiency for testing in pandemic scenarios, such as COVID-19? What are the possibilities and implications of using drones equipped with 5G technology for transporting medical supplies and samples?	To explore the potential of drones in enhancing the speed and efficiency of transporting testing supplies, such as swabs and reagents, during health crises like pandemics To investigate the feasibility and benefits of using drones with 5G technology for medical logistics, examining aspects like data transmission, real-time tracking, and route optimization

Second, the review illuminates the theoretical underpinnings driving current research, such as the use of optimization models in route planning, behavioral theories in understanding consumer acceptance, and risk management theories in public acceptance of drones. This diverse theoretical grounding hints at the interdisciplinary nature of drone research, encouraging researchers to borrow and adapt theories from various disciplines to create more robust and comprehensive theoretical frameworks. Third, the review brings attention to the dynamic interplay between technology, society, and regulations within the context of drone applications. The emerging themes of public acceptance, policy and regulatory frameworks, and societal implications underscore the need for theories that can accommodate these multifaceted interactions and call for more integrative theoretical models. Finally, by delineating the future research agenda, the review sets the stage for theoretical advancements in underexplored areas. We emphasize the need for theories that address the challenges of integrating drones into urban operations, understand the barriers to drone adoption in supply chains, and explore the impact of drones on consumer behavior in food logistics, among others. This research agenda serves as a catalyst for researchers to push the boundaries of existing theories and develop new ones that can better explain, predict, and optimize the applications and implications of drones.

This article also unfolds numerous practical implications that can benefit various stakeholders, including policymakers, industry leaders, logistics providers, and consumers. As such, the review's comprehensive understanding of the existing drone applications provides industry practitioners with an overview of the current state of drone technology. It enables them to evaluate their own operations and practices against the industry's latest developments, helping them identify areas for potential adoption or improvement. Moreover, the review highlights the importance of understanding consumer behavior and public acceptance of drone technology. This insight is crucial for businesses looking to implement drone services as it underscores the need to address consumer concerns regarding safety, privacy, and

efficiency to ensure successful adoption. In addition, the challenges and issues related to drone delivery route optimization and logistical problem-solving can provide practical guidance for logistics providers and supply chain managers. The optimization models and strategies outlined in the review can be applied to enhance operational efficiency, reduce costs, and improve service quality. For policymakers, the identification of the regulatory and policy implications of drone technology provides valuable insights. The highlighted need for comprehensive and adaptive regulatory frameworks can guide the development of policies that balance technological innovation, safety, privacy, and public acceptance. Furthermore, the review's emphasis on the role of drones in medical and laboratory transport can inform healthcare providers and emergency services about the potential of drones to enhance healthcare delivery, especially in remote or difficult-to-reach areas. Finally, by outlining a future research agenda, the review provides practical guidance for research and development efforts in the drone industry. It points out areas where further technological advancements are needed and where new drone applications could emerge, thereby directing resources toward the most promising and impactful areas.

5.4 Limitations

While this systematic review provides valuable insights into the use of drones in transportation, logistics, and supply chain management, it is not without limitations. Firstly, the review is based on a dataset of 96 selected journal articles, curated through a specific set of criteria defined by the authors. While this process is meticulous, it might introduce certain biases, potentially leading to overrepresentation or underrepresentation of certain perspectives or areas of research. Furthermore, the conclusions drawn are primarily based on the analyzed dataset of journal articles. As a result, the applicability and generalizability of these findings to real-world drone applications in logistics and supply chain management could be limited. The omission of insights from other sources, such as books, conference papers, technical reports, and industry case studies, due to these selection criteria might further constrain the scope of our findings.

Secondly, the study utilizes LDA for topic modeling, an effective method for summarizing large volumes of text. Nevertheless, LDA's performance is contingent upon various factors such as the quality of the dataset, the preprocessing methods employed, and the selected number of topics. This reliance may influence the findings and thematic landscape of the paper as it may not perfectly capture the nuanced themes and subtleties inherent in the studies. Also, the current methodology might not fully utilize the potential of recent advances in techniques applied to process and analyze the images captured by drones [76, 77]. Therefore, to address these limitations, future research could consider a more diverse range of sources and a more comprehensive set of selection criteria to augment the dataset. It could also benefit from exploring complementary analytical methods or state-of-the-art techniques in order to broaden the scope and enhance the depth of the analysis.

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Data Availability Data are available upon request.

Declarations

Conflict of interest None.

References

1. Rejeb, A.; Rejeb, K.; Simske, S.J.; Treiblmaier, H.: Drones for supply chain management and logistics: a review and research agenda. Int. J. Logist. Res. Appl. (2021). https://doi.org/10.1080/13675567.2021.1981273
2. Aggarwal, A.K.: Fusion and Enhancement Techniques for Processing of Multispectral Images. In: Avtar, R.; Watanabe, T. (Eds.) Unmanned Aerial Vehicle: Applications in Agriculture and Environment, pp. 159–175. Springer International Publishing, Cham (2020)
3. Lemardelé, C.; Estrada, M.; Pagès, L.; Bachofner, M.: Potentialities of drones and ground autonomous delivery devices for last-mile logistics. Transp. Res. Part E Logist. Transp. Rev. (2021). https://doi.org/10.1016/j.tre.2021.102325
4. Xiao, J.; Suab, S.A.; Chen, X.; Singh, C.K.; Singh, D.; Aggarwal, A.K.; Korom, A.; Widyatmanti, W.; Mollah, T.H.; Minh, H.V.T.; Khedher, K.M.; Avtar, R.: Enhancing assessment of corn growth performance using unmanned aerial vehicles (UAVs) and deep learning. Measurement **214**, 112764 (2023). https://doi.org/10.1016/j.measurement.2023.112764
5. Damoah, I.S.; Ayakwah, A.; Tingbani, I.: Artificial intelligence (AI)-enhanced medical drones in the healthcare supply chain (HSC) for sustainability development: A case study. J. Clean. Prod. (2021). https://doi.org/10.1016/j.jclepro.2021.129598
6. Mohammad, W.A.M.; NazihDiab, Y.; Elomri, A.; Triki, C.: Innovative solutions in last mile delivery: concepts, practices, challenges, and future directions. Supply Chain Forum (2023). https://doi.org/10.1080/16258312.2023.2173488
7. Rejeb, A.; Rejeb, K.; Simske, S.; Treiblmaier, H.: Humanitarian Drones: A Review and Research Agenda. Internet Things Neth (2021). https://doi.org/10.1016/j.iot.2021.100434
8. Moshref-Javadi, M.; Winkenbach, M.: Applications and research avenues for drone-based models in logistics: a classification and review. Expert Syst. Appl. **177**, 114854 (2021). https://doi.org/10.1016/j.eswa.2021.114854
9. Kellermann, R.; Biehle, T.; Fischer, L.: Drones for parcel and passenger transportation: a literature review. Transp. Res. Interdiscip. Perspect. **4**, 100088 (2020). https://doi.org/10.1016/j.trip.2019.100088
10. Chung, S.H.; Sah, B.; Lee, J.: Optimization for drone and drone-truck combined operations: a review of the state of the art and future directions. Comput. Oper. Res.. Oper. Res. **123**, 105004 (2020). https://doi.org/10.1016/j.cor.2020.105004

11. Merkert, R.; Bushell, J.: Managing the drone revolution: A systematic literature review into the current use of airborne drones and future strategic directions for their effective control. J. Air Transp. Manag.Manag. **89**, 101929 (2020). https://doi.org/10.1016/j.jairtraman.2020.101929
12. Abdelmaboud, A.: The internet of drones: requirements, taxonomy, recent advances, and challenges of research trends. Sensors. **21**, 5718 (2021). https://doi.org/10.3390/s21175718
13. Zhang, R.; Dou, L.; Xin, B.; Chen, C.; Deng, F.; Chen, J.: A review on the truck and drone cooperative delivery problem. Unmanned Syst. (2023). https://doi.org/10.1142/S2301385024300014
14. Madani, B.; Ndiaye, M.: Hybrid truck-drone delivery systems: a systematic literature review. IEEE Access. **10**, 92854–92878 (2022). https://doi.org/10.1109/ACCESS.2022.3202895
15. Benarbia, T.; Kyamakya, K.: A literature review of drone-based package delivery logistics systems and their implementation feasibility. Sustainability. **14**, 360 (2022). https://doi.org/10.3390/su14010360
16. Li, Y.; Liu, M.; Jiang, D.: Application of unmanned aerial vehicles in logistics: a literature review. Sustainability. **14**, 14473 (2022). https://doi.org/10.3390/su142114473
17. Rojas Viloria, D.; Solano-Charris, E.L.; Muñoz-Villamizar, A.; Montoya-Torres, J.R.: Unmanned aerial vehicles/drones in vehicle routing problems: a literature review. Int. Trans. Oper. Res.Oper. Res. **28**, 1626–1657 (2021). https://doi.org/10.1111/itor.12783
18. Kozlowski, D.; Semeshenko, V.; Molinari, A.: Latent Dirichlet allocation model for world trade analysis. PLoS ONE **16**, e0245393 (2021). https://doi.org/10.1371/journal.pone.0245393
19. Blei, D.M.; Ng, A.Y.; Jordan, M.I.: Latent dirichlet allocation. J. Mach. Learn. Res. **3**, 993–1022 (2003)
20. Blei, D.M.: Probabilistic topic models. Commun. ACM. ACM **55**, 77–84 (2012). https://doi.org/10.1145/2133806.2133826
21. Guo, Y.; Barnes, S.J.; Jia, Q.: Mining meaning from online ratings and reviews: tourist satisfaction analysis using latent dirichlet allocation. Tour. Manag.Manag. **59**, 467–483 (2017). https://doi.org/10.1016/j.tourman.2016.09.009
22. Moro, S.; Cortez, P.; Rita, P.: Business intelligence in banking: A literature analysis from 2002 to 2013 using text mining and latent Dirichlet allocation. Expert Syst. Appl. **42**, 1314–1324 (2015). https://doi.org/10.1016/j.eswa.2014.09.024
23. Abdollahi, A.; Rejeb, K.; Rejeb, A.; Mostafa, M.M.; Zailani, S.: Wireless sensor networks in agriculture: Insights from bibliometric analysis. Sustain. Switz. (2021). https://doi.org/10.3390/su132112011
24. Rejeb, A.; Suhaiza, Z.; Rejeb, K.; Seuring, S.; Treiblmaier, H.: The Internet of Things and the circular economy: a systematic literature review and research agenda. J. Clean. Prod. **350**, 131439 (2022). https://doi.org/10.1016/j.jclepro.2022.131439
25. McCallum, A.K.: Mallet: A machine learning for languagetoolkit. Httpmallet Cs Umass Edu. (2002)
26. Aria, M.; Cuccurullo, C.: Bibliometrix: an R-tool for comprehensive science mapping analysis. J. Informetr. **11**, 959–975 (2017). https://doi.org/10.1016/j.joi.2017.08.007
27. Caputo, A.; Pizzi, S.; Pellegrini, M.M.; Dabić, M.: Digitalization and business models: where are we going? A science map of the field. J. Bus. Res. **123**, 489–501 (2021). https://doi.org/10.1016/j.jbusres.2020.09.053
28. Tatham, P.: An investigation into the suitability of the use of unmanned aerial vehicle systems (UAVS) to support the initial needs assessment process in rapid onset humanitarian disasters. Int. J. Risk Assess. Manag. **13**, 60–78 (2009). https://doi.org/10.1504/IJRAM.2009.026391
29. Zhang, G.; Jia, N.; Zhu, N.; Adulyasak, Y.; Ma, S.: Robust drone selective routing in humanitarian transportation network assessment. Eur. J. Oper. Res.Oper. Res. **305**, 400–428 (2023). https://doi.org/10.1016/j.ejor.2022.05.046
30. Moshref-Javadi, M.; Lee, S.; Winkenbach, M.: Design and evaluation of a multi-trip delivery model with truck and drones. Transp. Res. Part E Logist. Transp. Rev. (2020). https://doi.org/10.1016/j.tre.2020.101887
31. Sievert, C., Shirley, K.: LDAvis: A method for visualizing and interpreting topics. In: Proceedings of the workshop on interactive language learning, visualization, and interfaces. pp. 63–70 (2014)
32. Asadi, A.; NurrePinkley, S.: A stochastic scheduling, allocation, and inventory replenishment problem for battery swap stations. Transp. Res. Part E Logist. Transp. Rev. (2021). https://doi.org/10.1016/j.tre.2020.102212
33. MbiadouSaleu, R.G.; Deroussi, L.; Feillet, D.; Grangeon, N.; Quilliot, A.: The parallel drone scheduling problem with multiple drones and vehicles. Eur. J. Oper. Res.Oper. Res. **300**, 571–589 (2022). https://doi.org/10.1016/j.ejor.2021.08.014
34. Wang, K.; Pesch, E.; Kress, D.; Fridman, I.; Boysen, N.: The piggyback transportation problem: transporting drones launched from a flying warehouse. Eur. J. Oper. Res.Oper. Res. **296**, 504–519 (2022). https://doi.org/10.1016/j.ejor.2021.03.064
35. Chowdhury, S.; Emelogu, A.; Marufuzzaman, M.; Nurre, S.G.; Bian, L.: Drones for disaster response and relief operations: A continuous approximation model. Int. J. Prod. Econ. **188**, 167–184 (2017). https://doi.org/10.1016/j.ijpe.2017.03.024
36. de Freitas, J.C.; Penna, P.H.V.: A variable neighborhood search for flying sidekick traveling salesman problem. Int. Trans. Oper. Res.Oper. Res. **27**, 267–290 (2020). https://doi.org/10.1111/itor.12671
37. Demir, E.; Syntetos, A.; Van Woensel, T.: Last mile logistics: Research trends and needs. IMA J. Manag. Math.Manag. Math. **33**, 549–561 (2022). https://doi.org/10.1093/imaman/dpac006
38. Ali, S.S.; Khan, S.; Fatma, N.; Ozel, C.; Hussain, A.: Utilisation of drones in achieving various applications in smart warehouse management. Benchmarking. (2023). https://doi.org/10.1108/BIJ-01-2023-0039
39. Santos, R.; de Melo, R.; Bastos Costa, D.: Integrating resilience engineering and UAS technology into construction safety planning and control. Eng. Constr. Archit. Manag.Manag. **26**, 2705–2722 (2019). https://doi.org/10.1108/ECAM-12-2018-0541
40. Zhang, J.Z.; Srivastava, P.R.; Eachempati, P.: Evaluating the effectiveness of drones in emergency situations: a hybrid multi-criteria approach. Ind. Manag. Data Syst.Manag. Data Syst. **123**, 302–323 (2023). https://doi.org/10.1108/IMDS-01-2021-0064
41. Akbari, M.; Hopkins, J.L.: Digital technologies as enablers of supply chain sustainability in an emerging economy. Oper. Manag. Res.. Manag. Res. **15**, 689–710 (2022). https://doi.org/10.1007/s12063-021-00226-8
42. Jeong, H.Y.; Yu, D.J.; Min, B.-C.; Lee, S.: The humanitarian flying warehouse. Transp. Res. Part E Logist. Transp. Rev. (2020). https://doi.org/10.1016/j.tre.2020.101901
43. Akhtar, P.; Osburg, V.-S.; Kabra, G.; Ullah, S.; Shabbir, H.; Kumari, S.: Coordination and collaboration for humanitarian operational excellence: big data and modern information processing systems. Prod. Plan. Control. **33**, 705–721 (2022). https://doi.org/10.1080/09537287.2020.1834126
44. Tatham, P.; Stadler, F.; Murray, A.; Shaban, R.Z.: Flying maggots: a smart logistic solution to an enduring medical challenge. J. Humanit. Logist. Supply Chain Manag. **7**, 172–193 (2017). https://doi.org/10.1108/JHLSCM-02-2017-0003
45. Rave, A.; Fontaine, P.; Kuhn, H.: Drone location and vehicle fleet planning with trucks and aerial drones. Eur. J. Oper. Res.Oper. Res. **308**, 113–130 (2023). https://doi.org/10.1016/j.ejor.2022.10.015
46. Amorosi, L.; Puerto, J.; Valverde, C.: An extended model of coordination of an all-terrain vehicle and a multivisit drone. Int. Trans. Oper. Res.Oper. Res. (2022). https://doi.org/10.1111/itor.13179
47. Vu, L.; Vu, D.M.; Minh Hoàng, H.À.; Nguyen, V.-P.: The two-echelon routing problem with truck and drones. Int. Trans. Oper.

Res.Oper. Res. **29**, 2968–2994 (2022). https://doi.org/10.1111/itor.13052
48. Salama, M.R.; Srinivas, S.: Collaborative truck multi-drone routing and scheduling problem: package delivery with flexible launch and recovery sites. Transp. Res. Part E Logist. Transp. Rev. (2022). https://doi.org/10.1016/j.tre.2022.102788
49. Kamat, A.; Shanker, S.; Barve, A.: Assessing the factors affecting implementation of unmanned aerial vehicles in Indian humanitarian logistics: a g-DANP approach. J. Model. Manag.Manag. **18**, 416–456 (2023). https://doi.org/10.1108/JM2-02-2021-0037
50. Kamat, A.; Shanker, S.; Barve, A.; Muduli, K.; Mangla, S.K.; Luthra, S.: Uncovering interrelationships between barriers to unmanned aerial vehicles in humanitarian logistics. Oper. Manag. Res.. Manag. Res. **15**, 1134–1160 (2022). https://doi.org/10.1007/s12063-021-00235-7
51. Sah, B.; Gupta, R.; Bani-Hani, D.: Analysis of barriers to implement drone logistics. Int. J. Logist. Res. Appl. **24**, 531–550 (2021). https://doi.org/10.1080/13675567.2020.1782862
52. Baniasadi, P.; Foumani, M.; Smith-Miles, K.; Ejov, V.: A transformation technique for the clustered generalized traveling salesman problem with applications to logistics. Eur. J. Oper. Res.Oper. Res. **285**, 444–457 (2020). https://doi.org/10.1016/j.ejor.2020.01.053
53. Pei, Z.; Dai, X.; Yuan, Y.; Du, R.; Liu, C.: Managing price and fleet size for courier service with shared drones. Omega U.K. (2021). https://doi.org/10.1016/j.omega.2021.102482
54. Carlsson, J.G.; Song, S.: Coordinated logistics with a truck and a drone. Manag. Sci.. Sci. **64**, 4052–4069 (2018). https://doi.org/10.1287/mnsc.2017.2824
55. Irizarry, J.; Costa, D.B.: Exploratory study of potential applications of unmanned aerial systems for construction management tasks. J. Manag. Eng.Manag. Eng. (2016). https://doi.org/10.1061/(ASCE)ME.1943-5479.0000422
56. Li, Y.; Liu, C.: Applications of multirotor drone technologies in construction management. Int. J. Constr. Manag.Manag. **19**, 401–412 (2019). https://doi.org/10.1080/15623599.2018.1452101
57. Chen, X.; Chang-Richards, A.Y.; Pelosi, A.; Jia, Y.; Shen, X.; Siddiqui, M.K.; Yang, N.: Implementation of technologies in the construction industry: a systematic review. Eng. Constr. Archit. Manag.Manag. **29**, 3181–3209 (2022). https://doi.org/10.1108/ECAM-02-2021-0172
58. Appelbaum, D.; Nehmer, R.A.: Using drones in internal and external audits: An exploratory framework. J. Emerg. Technol. Account. **14**, 99–113 (2017). https://doi.org/10.2308/jeta-51704
59. Christ, M.H.; Emett, S.A.; Summers, S.L.; Wood, D.A.: Prepare for takeoff: improving asset measurement and audit quality with drone-enabled inventory audit procedures. Rev. Account. Stud. **26**, 1323–1343 (2021). https://doi.org/10.1007/s11142-020-09574-5
60. Ali, S.S.; Kaur, R.; Khan, S.: Identification of innovative technology enablers and drone technology determinants adoption: a graph theory matrix analysis framework. Oper. Manag. Res.. Manag. Res. (2023). https://doi.org/10.1007/s12063-023-00346-3
61. Rathore, B.; Gupta, R.; Biswas, B.; Srivastava, A.; Gupta, S.: Identification and analysis of adoption barriers of disruptive technologies in the logistics industry. Int. J. Logist. Manag.Logist. Manag. **33**, 136–169 (2022). https://doi.org/10.1108/IJLM-07-2021-0352
62. Mohamed, N.; Al-Jaroodi, J.; Jawhar, I.; Idries, A.; Mohammed, F.: Unmanned aerial vehicles applications in future smart cities. Technol. Forecast. Soc. Change. (2020). https://doi.org/10.1016/j.techfore.2018.05.004
63. Bubelíny, O.; Ďaďová, I.; Kubina, M.; Soviar, J.: The use of smart elements for the transport operation in the slovak cities. LOGI - Sci. J. Transp. Logist. **10**, 51–60 (2019). https://doi.org/10.2478/logi-2019-0015
64. Khalil, A.; Shankar, A.; Bodhi, R.; Behl, A.; Ferraris, A.: Why do people resist drone food delivery services? An innovation resistance theory perspective. IEEE Trans. Eng. Manag.Manag. (2022). https://doi.org/10.1109/TEM.2022.3202485
65. Pinto, R.; Zambetti, M.; Lagorio, A.; Pirola, F.: A network design model for a meal delivery service using drones. Int. J. Logist. Res. Appl. **23**, 354–374 (2020). https://doi.org/10.1080/13675567.2019.1696290
66. Bafouni-Kotta, A.; Villanueva, J.; Kalakou, S.; Dias, J.G.: Upcoming drone delivery services: a perceptions-based clustering of citizens in Lisbon. Technol. Soc. (2023). https://doi.org/10.1016/j.techsoc.2023.102250
67. Cai, L.; Yuen, K.F.; Xie, D.; Fang, M.; Wang, X.: Consumer's usage of logistics technologies: integration of habit into the unified theory of acceptance and use of technology. Technol. Soc. (2021). https://doi.org/10.1016/j.techsoc.2021.101789
68. Merkert, R.; Bliemer, M.C.J.; Fayyaz, M.: Consumer preferences for innovative and traditional last-mile parcel delivery. Int. J. Phys. Distrib. Logist. Manag.Distrib. Logist. Manag. **52**, 261–284 (2022). https://doi.org/10.1108/IJPDLM-01-2021-0013
69. Zhu, X.; Pasch, T.J.; Bergstrom, A.: Understanding the structure of risk belief systems concerning drone delivery: a network analysis. Technol. Soc. (2020). https://doi.org/10.1016/j.techsoc.2020.101262
70. Kellermann, R.; Fischer, L.: Drones for parcel and passenger transport: A qualitative exploration of public acceptance [Drones para el transporte de paquetes y pasajeros: Un análisis cualitativo de la aceptación pública]. Sociol. Tecnociencia. **10**, 106–138 (2020). https://doi.org/10.24197/st.2.2020.106-138
71. Smith, A.; Dickinson, J.E.; Marsden, G.; Cherrett, T.; Oakey, A.; Grote, M.: Public acceptance of the use of drones for logistics: the state of play and moving towards more informed debate. Technol. Soc. (2022). https://doi.org/10.1016/j.techsoc.2022.101883
72. Colajanni, G.; Daniele, P.; Sciacca, D.: Reagents and swab tests during the COVID-19 Pandemic: an optimized supply chain management with UAVs. Oper. Res. Perspect. (2022). https://doi.org/10.1016/j.orp.2022.100257
73. Koshta, N.; Devi, Y.; Patra, S.: Aerial bots in the supply chain: a new ally to combat COVID-19. Technol. Soc. (2021). https://doi.org/10.1016/j.techsoc.2021.101646
74. Nyaaba, A.A.; Ayamga, M.: Intricacies of medical drones in healthcare delivery: Implications for Africa. Technol. Soc. (2021). https://doi.org/10.1016/j.techsoc.2021.101624
75. Pulver, A.; Wei, R.: Optimizing the spatial location of medical drones. Appl. Geogr.Geogr. **90**, 9–16 (2018). https://doi.org/10.1016/j.apgeog.2017.11.009
76. Garg, M.; Ubhi, J.S.; Aggarwal, A.K.: Neural style transfer for image steganography and destylization with supervised image to image translation. Multimed. Tools Appl. **82**, 6271–6288 (2023). https://doi.org/10.1007/s11042-022-13596-3
77. Ubhi, J.S.; Aggarwal, A.K.: Neural Style Transfer for image within images and conditional GANs for destylization. J. Vis. Commun. Image Represent.Commun. Image Represent. **85**, 103483 (2022). https://doi.org/10.1016/j.jvcir.2022.103483

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